

Pathomporn Niamraksa

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SKILLS

Programming Languages : C/C++ (OpenGL, Vulkan), Python (NumPy, Matplotlib, OpenGL, multiprocessing)

Development & Build Tools : Visual Studio Code, Git, CMake, GCC

Operating System : Windows, Linux

Other Tools : LaTeX, Google Docs, Google Sheets, Google Slides, GIMP (Basic), Inkscape (Basic), TeamViewer, Canva (Basic)
Oracle VirtualBox (Basic)

EDUCATION

Ramkhamhaeng University

January 2026

Bachelor of Science in Physics : Cumulative GPA 2.08

- **Physics** : Classical Mechanics, Quantum Mechanics, Computational Physics
- **Computer Science** : Object-Oriented Programming, Data Structures, Algorithms, Numerical Methods, Computer Graphics

PROFESSIONAL EXPERIENCE

ClickRobot Rama 9

February - April 2026

Teacher

- Teaching students using **LEGO Essential**, **LEGO WeDo 2.0**, **LEGO Spike Prime**, **LEGO Mindstorms EV3** and **VEX IQ**
- **Scratch Programming** covering essential logic such as **Movements**, **Sensors**, **Events**, **Control**, **Operators**, **Variables** and **My Blocks**
- **Python Programming** covering core concepts such as **Syntax**, **Variables**, **Data Types**, **Operators**, **Conditionals (If-Else)**, **Loops (For/While)** and **Functions** applied to **LEGO Spike Prime** and **VEX IQ**

PROJECTS

Senior Project : Simulation of 2D Brownian Motion and the Fluctuation Theorem in Optical Tweezers

<https://github.com/YukiSnowy/BM-and-FT-in-OT>

- Developed a numerical simulation to model stochastic particle dynamics in an optical trap, solving the Langevin equation using Python.
- Demonstrated the fluctuation theorem by statistical analysis of entropy production in non-equilibrium Brownian dynamics.
- Implemented real-time 2D visualization of stochastic Brownian particle trajectories in an optical trap with Periodic boundary conditions.

RU-Thesis : LaTeX Thesis Template for Ramkhamhaeng University

<https://github.com/YukiSnowy/RU-Thesis>

- Developed a custom LaTeX class for university thesis formatting.
- Designed to be easy to use and user-friendly.
- Supports multiple languages, including Thai and English.